Data Governance, Compliance and Ethics

ETHICAL ISSUES PERTAINING TO DATA (PART 3)



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Overview of course content

1. Data Governance and Management (Weeks 1-3)
2. Privacy and Data Protection (Weeks 4-6)
3. Ethical Issues Pertaining to Data (Weeks 7-9)
4. Fairness, Accountability, and Transparency of Algorithmic Systems (Weeks 10,11)

Ethical Issues Pertaining to Data

Ethics and Computing – examining moral problems when using the Internet - spam, censorship and free speech, anonymity offered by the Internet.

Ethical issues arising from the increasing use and pervasiveness of Information Technology and socio-technical systems.

Health technology.. Pervasive monitoring and tracking.. Image, video and sound capture.. Identity.. Perpetuity of data storage.. Transnationality..

Ethical Issues Pertaining to Data (Part 3)

Health Technology – Ethical aspects



Definition of eHealth (integration of technology in delivering health services)

"eHealth (Electronic Health) involves the integration of all information and knowledge sources involved in the delivery of healthcare via information technology-based systems.

This includes patients and their records, caregivers and their systems, monitoring devices and sensors, management and administrative functions.

It is a fully integrated digital 'supply chain' and involves high levels of automation and information sharing."

From eHealth-Strategy-for-Ireland.pdf (link on Moodle)

The case for eHealth

"eHealth can benefit citizens, patients, health and care professionals but also health organisations and public authorities. eHealth - when applied effectively - delivers more personalised 'citizen-centric' healthcare, which is more targeted, effective and efficient and helps reduce errors, as well as the length of hospitalisation.

It facilitates socio-economic inclusion and equality, quality of life and patient empowerment through greater transparency, access to services and information and the use of social media for health"

European Union eHealth Action Plan 2012-2020.

eHealth Potential priority projects (Ireland)

National Health Identifier Infrastructure.

ePrescribing Systems.

Online Referrals and Scheduling.

Tele-healthcare - particularly relating to the management of chronic diseases.

Development of Patient Summary Records.

Online Access to Health Information.

National Patient Portal.

An Electronic Health Record as a key part of eHealth

An electronic health record (EHR) is a record of a patient's medical details (including history, physical examination, investigations and treatment) in digital format.

Physicians and hospitals are implementing EHRs because:

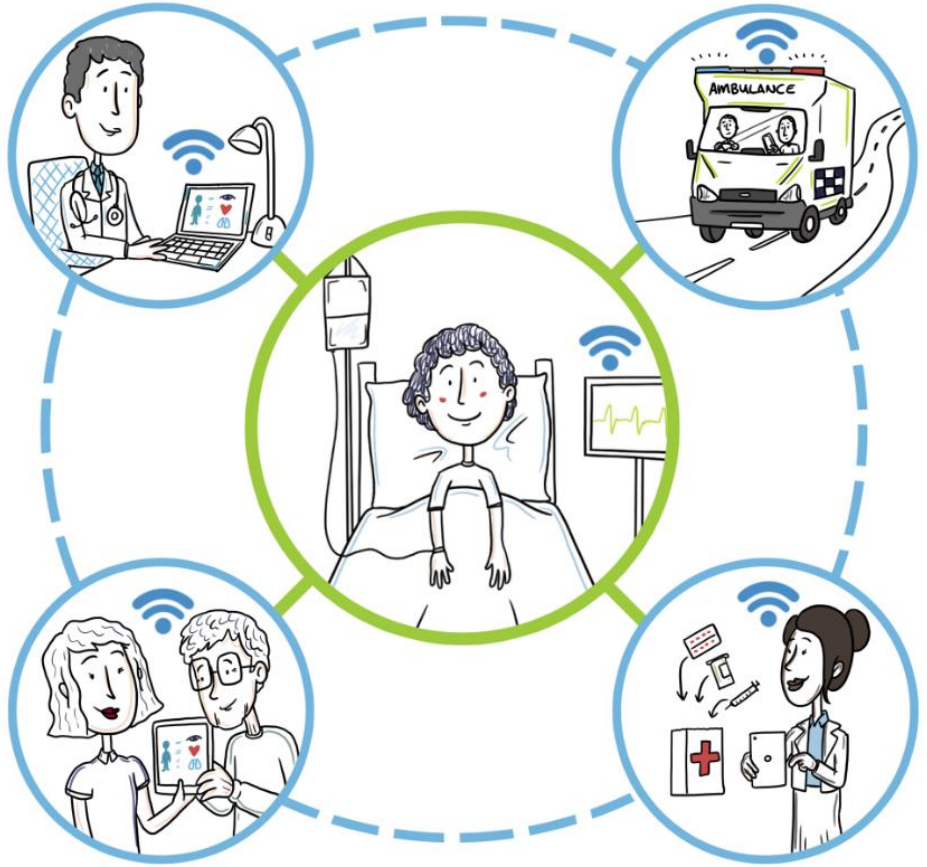
- increase access to health care
- improve collaboration
- improve the quality of care
- enable fast access to services
- decrease costs



https://youtu.be/t8a7XO7MuQE

Potential Benefits of EHR

1. Production of legible records reduces many problems of wrong prescriptions, doses and procedure
2. Easy accessibility from anywhere at any given time
3. Require less storage space and can be stored indefinitely
4. Reduce the number of lost records
5. Help research activities
6. Allow for a complete set of backup records at low cost
7. Speed data transfer
8. Cost-effective



Addressing concerns linked to EHRS

Despite potential benefits, there are concerns

1. Privacy and confidentiality/Data ownership,
2. Risk of security breaches
3. System implementation
4. Data quality and risk of inaccuracies



1. Protect Privacy And Confidentiality

Privacy *is the right of an individual to keep information about themselves from being disclosed to others; the claim of individuals to be let alone, from surveillance or interference from other individuals, organizations or the government.*



Requires:

- authorizing users - allow only authorized individuals to have access to information
- user's access is based on preestablished role-based privileges
- strong privacy and security policies

2. Minimise Risk of Security Breaches

Security breaches threaten patient privacy when confidential health information is made available to others without the individual's consent or authorization. Measures to make EHRs more secure could include:



Secure storage and access

- Cloud storage, password protection, and encryption
- Usage of two-factor authentication system with security tokens and password
- Firewalls, antivirus software, and intrusion detection software

Robust Policies and Procedures

- Specific policies and procedures serve to maintain patient privacy and confidentiality employees must not share their ID with anyone, always log off when leaving a terminal and use their own ID to access patient digital records.

Clear roles and responsibilities

- A security officer must be designated by the organization to work with a team of health IT experts

Effective Auditing procedures

- Routine random audits conducted on a regular basis to ensure compliance with hospital policy
- All system activity can be tracked by audit trails

3. Secure system Implementation

There are different **challenges in the implementation of EHR:**

- Lack of involvement from appropriate stakeholders
- Clinical personnel often have little knowledge of the clinic's workflow and the roles others play in care delivery
- Clinics should map and standardize their workflows before EHR selection
- Improperly designed user interface lead to deliver poor quality of care
- Maintenance and testing of these interfaces on a routine basis is essential in controlling this major risk.



4. Improve Data quality and reduce risk of inaccuracies

Integrity assures that the data is accurate and has not been changed

Some aspects that may cause inaccuracies:

- Improper use of “cut and paste” option
- Drop down menus limit the choices available to the clinician
- Medical identity theft



A common health identifier for Ireland - Discussion

Health systems presents a number of arguments about why a single health identifier is critical.
Do you agree?



What protections would you put in place to protect against misuse or to improve public confidence in its rollout?



It is critical for the **provision of eHealth systems** for:

- Identifying individual patients in health care systems.
- Identifying individual's health records in health care systems.
- Linkage of health records across systems for presentation to health care providers.
- Electronic recording of patient transfers and referrals across the health systems, across all health domains (acute, community, disease registers, public and private, national and international).
- Complying with GDPR, supporting audit requirements on health record access logs.

Source: https://www.ehealthireland.ie/A2I-HIDs-Programme/Individual-Health-Identifier-IHI-/ (accessed 27/10/2020)

Pervasive Monitoring And Tracking using (H-IoT) technologies

Health-related internet of things (H-IoT) technologies increasingly play a key role in health management:

- Disease Prevention
- Real-time tele-monitoring of patient's functions
- Testing of treatments
- Fitness And well-being Monitoring
- Medication Dispensation
- Health Research Data Collection

However, it also raises ethical problems from the risks of Internet enabled devices, the sensitivity of health-related data, and their impact on the delivery of healthcare.

Source: Ethics of the health-related internet of things: a narrative review

Potential benefits of H-IOT tech

Increased access to care especially where currently limited

Improved patient safety and engagement from better customer information and feedback.

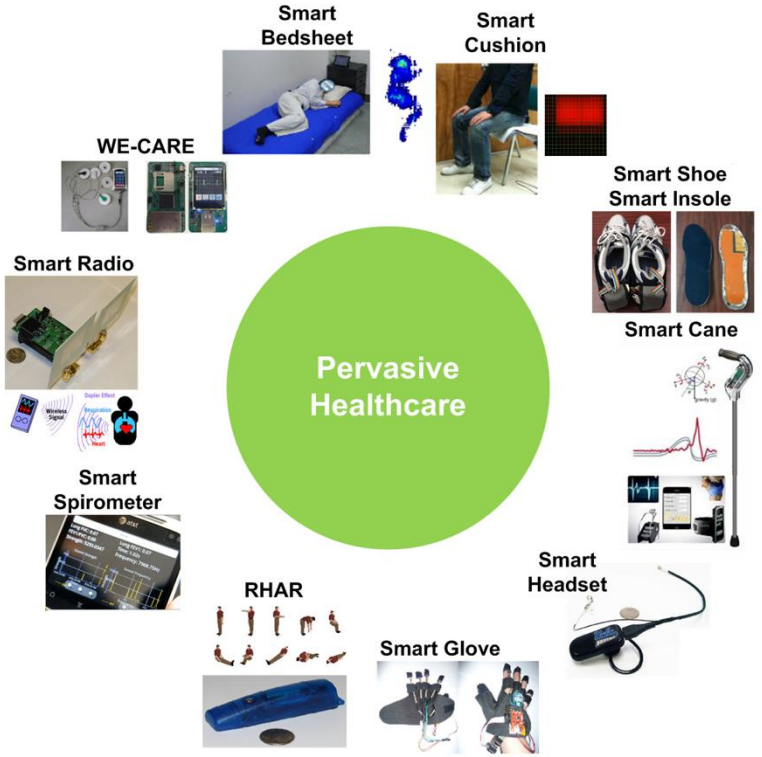
More effective medical research from longitudinal and more granular data

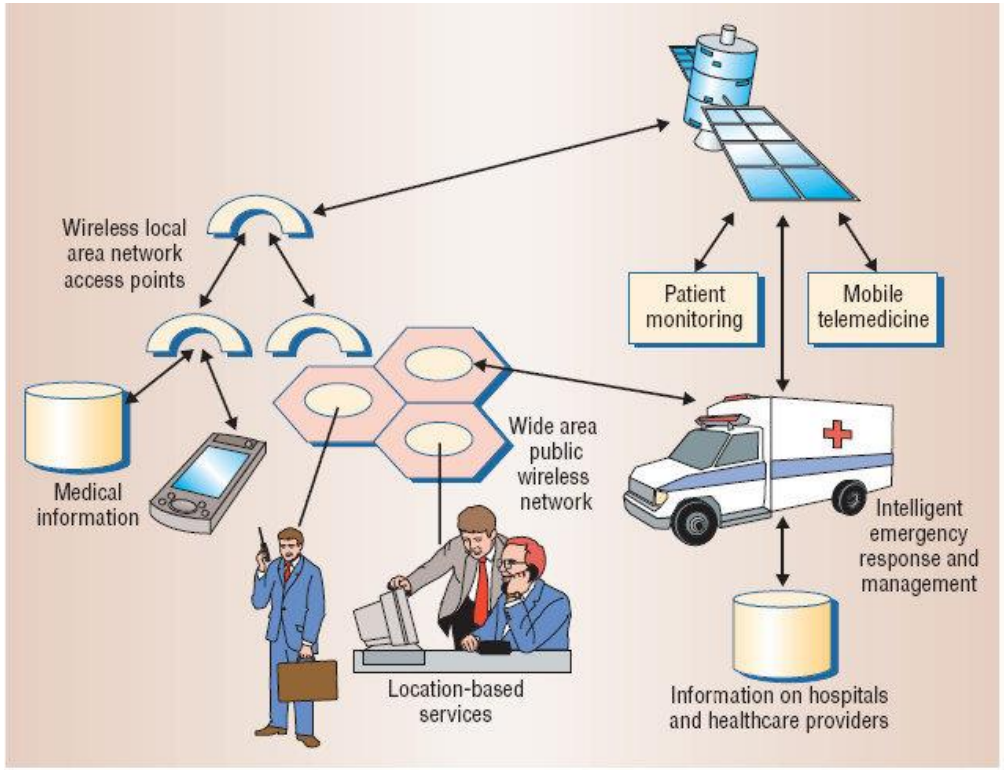


Pervasive Monitoring And Tracking – (H-IoT) technologies

Pervasive Healthcare Systems use large-scale deployment of sensing and communication technologies

Their aim is to continuously monitor patients' history and subsequently to deliver accurate health information to medical professionals





Pervasive Monitoring And Tracking – (H-IoT) technologies

Ethical issues/concerns for H-IoT technologies can be considered from a number of perspectives:

1. Devices used
2. Data protocols observed
3. Medical and social care practice levels



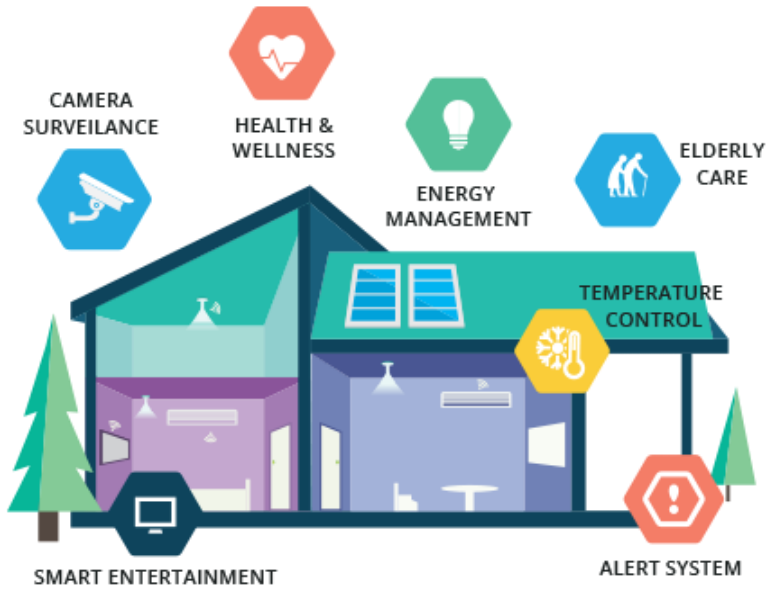
Ethical issues for H-IoT devices (1. Devices used)

Personal privacy needs

- right to possess and protect personal space
- the right to be left alone or not monitored by a third party
- freedom to “escape being observed or accessed when desired”
- Distinction between commercial health trackers (eg fitbits) and clinical devices

Obtrusiveness, stigma and autonomy

- The technology that are perceived as undesirable and physically or psychologically prominent
- While sensors ‘fading into the background’ may make users more comfortable, the psychological disappearance of H-IoT when used in personal spaces can also cause ethical problems
- Validity of consent may require ongoing review and renewal

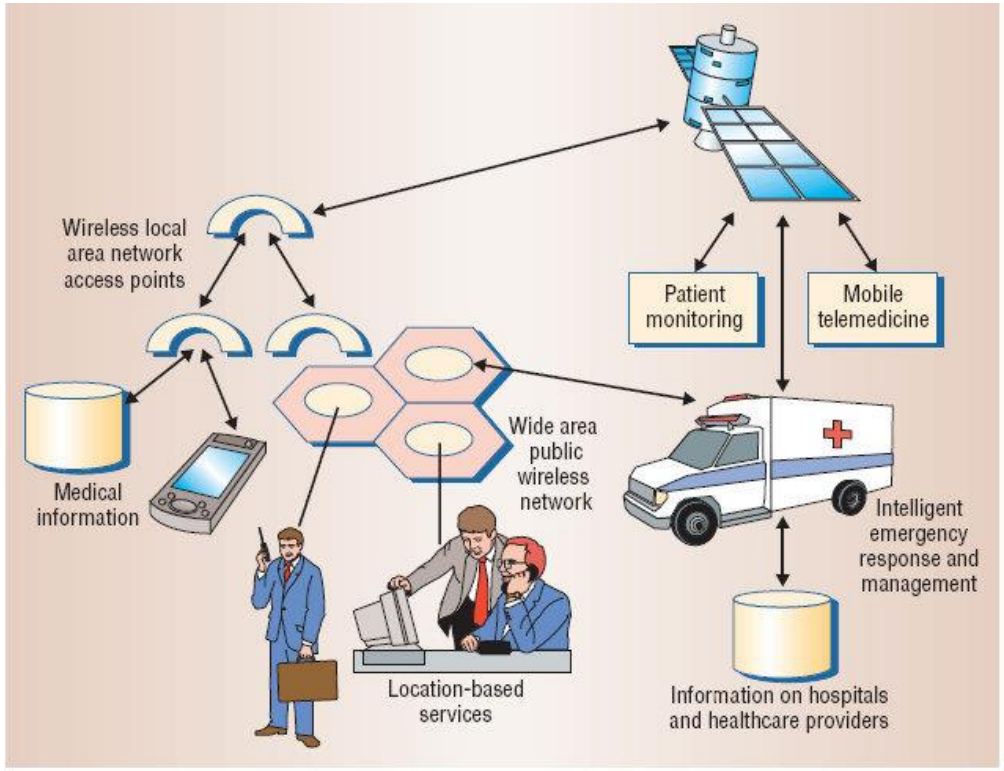


Pervasive Monitoring And Tracking – (H-IoT) technologies

Different from a traditional paper-based system, a pervasive healthcare system collects and manages health data in an electronic record. This means that a significant portion of information, much of which is sensitive, about patients 'health status (e.g., blood pressure and heartbeat) is collected from different devices monitoring the patients' health and recorded in electronic format for subsequent processing.

These devices also monitor the processed information (can also be sent from other such devices) and alarm the medical staff only whenever like, for example, blood pressure be-comes out of the expected range for some specific patient. These all are done without necessarily requiring intervention by the medical staff(s).

However, such usage imposes many security risks to the health information that did not exist before with paper-based system.



Ethical issues for H-IoT devices (2. Data Protocols)

Informational privacy (contrasts with personal privacy)

- To gain control over the spread of information about the user's health status and history
- Local anonymization or allowing users to enforce privacy preferences can help to prevent unauthorized access or identification of users



Data sharing and autonomy

- Autonomy refers to freedom and independence in reference to assistive technologies
- The freedom of users may be impeded due to the presence of sensors or transmission of data generated by H-IoT devices

Consent and the uncertain value of H-IoT data

- Traditional models of informed consent may not be directly applicable to H-IoT data
- Consent is normally granted for participation in a single study, not covering unrelated investigations resulting from data

Ethical issues for H-IoT devices (3. medical and social care practice levels)



Social isolation

- H-IoT will replace personal and social interactions with carers rather than merely supplementing them
- Collection of contextual information about a patient's condition via face-to-face interactions can be difficult to replicate with sensors

Decontextualization of health and well-being

- H-IoT can limit assessment of a patient's condition to a narrow range of easily measurable or quantifiable considerations, which could bias assessment

"Good" care and user well being

- H-IoT can produce a poorer view of social and contextual factors of a patient's health
- Care providers may be less able to demonstrate understanding and compassion to the patient's case
- Risk of increasing reliance on non-professional care

Ethical challenges of digital identity



The **definition of identity** is who you are, the way you think about yourself, the way you are viewed by the world and the characteristics that **define** you. An example of **identity** is a person's name .

The ethical challenges of digital identity

- the growing volume of information collected makes the notion of systematic user consent and control unrealistic
- changes in the nature of technical collection methods, as demonstrated by the advent of connected objects, has led to the increase of sensors collecting data even without the user realizing
- the treatment of human beings as though they were inanimate objects

Identity - Pervasive monitoring and tracking

A digital identify allows location and activities to be monitored

This can have some significant advantages – what are they?

It is still challenging someone's privacy, potentially violating individual rights through misuse of the information or use for cynical purpose

It means individual's privacy is not respected, private, confidential, secure and controlled. Hence, ethical issues arises.



Surveillance



Unexpected Uses of Data



Data Quality

Monitoring activity and location

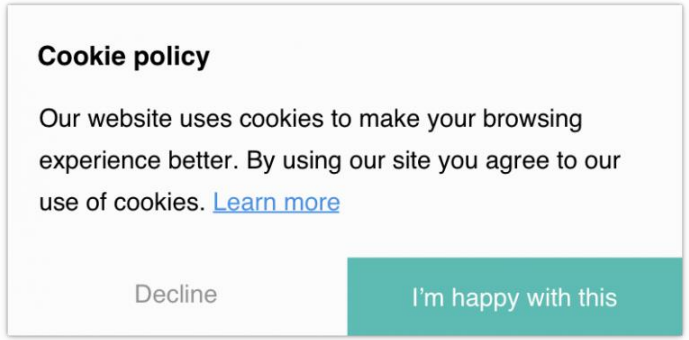
Cookies, web bugs, spyware, internet protocol (IP), loyalty cards are tools being used to monitor, track and store personal information, activities and users' behaviour online.

The collected data is mainly utilized for market analysis, direct marketing or advertisement.

What are the ethical issues implied here?

What's ethical and unethical about monitoring depends on the situation

Is it ethical for companies to buy our information in exchange for services?







GPS and location tracking

Global Positioning System (GPS) is a series of satellites orbiting around the Earth that provide real-time information on a location of a GPS receiver.

Many GPS systems are a way to help track individuals who **need to be monitored.**

GPS-based movement tracking presents a classic example of the 'double-edged sword' with respect to its use in health research and ethical concerns

Measures to address ethical concerns:

- Anonymization and de-identification
- Simple and clear explanations during informed consent
- Selective archival of data
- Secure storage



A question arises as to who actually owns the data produced through the system.

Anonymity

- Anonymity can be an asset to protect, secure privacy and freedom of expression
- Anonymity has benefits to cybercriminals
- How should one know, if a person's decision to remain anonymous is justifiable?
- It is not ethically acceptable for a person to hide their true identity
- Discrepancy of our online behavior compared to our real life can be considered unethical

Influencers react to Norway photo edit law: 'Welcome honesty' or a 'shortcut'?

By Kirsty Grant
Newsbeat reporter

A new law in Norway is coming into force that will mean social media influencers can't post modified photos without declaring what they've done.

The rules will affect any paid posts across all social media platforms, as part of an effort to "reduce body pressure" among young people.

Influencers react to Norway photo edit law: 'Welcome honesty' or a 'shortcut'? - BBC News

Image, video and sound capture - issues



Image Theft

- Using other people's work (photograph, design) for their own benefit
- Just copied or manipulated



Photo Plagiarism

- a photographer borrows another photographer's idea and perspective to get a very similar, if not identical shot
- Is it acceptable to re-capture such images, taking inspiration from the original author?



Photo: https://petapixel.com/2017/04/01/plagiarism-coincidence-curious-case-sony-world-photo-awards/

Image, video and sound capture - issues

Photo Manipulation

- How much manipulation should be allowed and what are the limits?

Drone Photography

- Drone photography is something very new and largely unregulated
- a number of cases of drones posing threat to passenger and government aircraft, falling into protected areas or crashing into buildings
- flying over private property, privacy concerns and recording footage of people or their property



Image, video and sound capture - issues

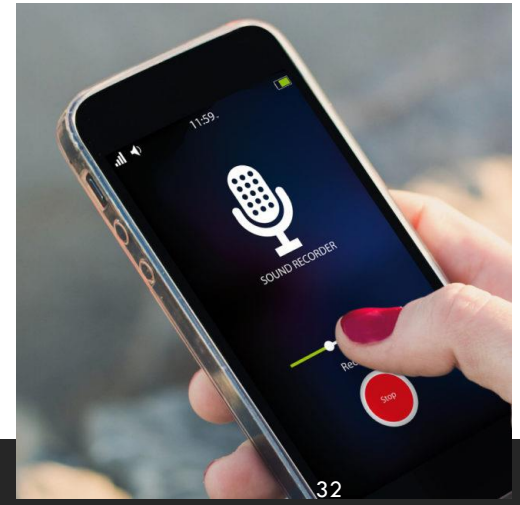
Ethical issues in sound capture - *what's acceptable and what's not when it comes to editing audio for journalistic purposes?*

If you cover an outdoor protest where demonstrators are singing John Lennon's Imagine, but a windy day created some mike noise, can you just substitute the sound from another demonstration the previous week where different protestors sang the same song?

if you interview a carpenter but fail to record the sound of him at work in his workshop, can you just record yourself using a hammer at home later and pretend, in your report, that it is the sound of the carpenter at work.

You can't do either of these things and honour the first principle of journalism – **accuracy**

- NEVER use sound you did not record yourself or fabricated sound
- NEVER change the meaning of what the interviewee said
- The question of the ethical acceptability of recording in public places is a grey, and changing area

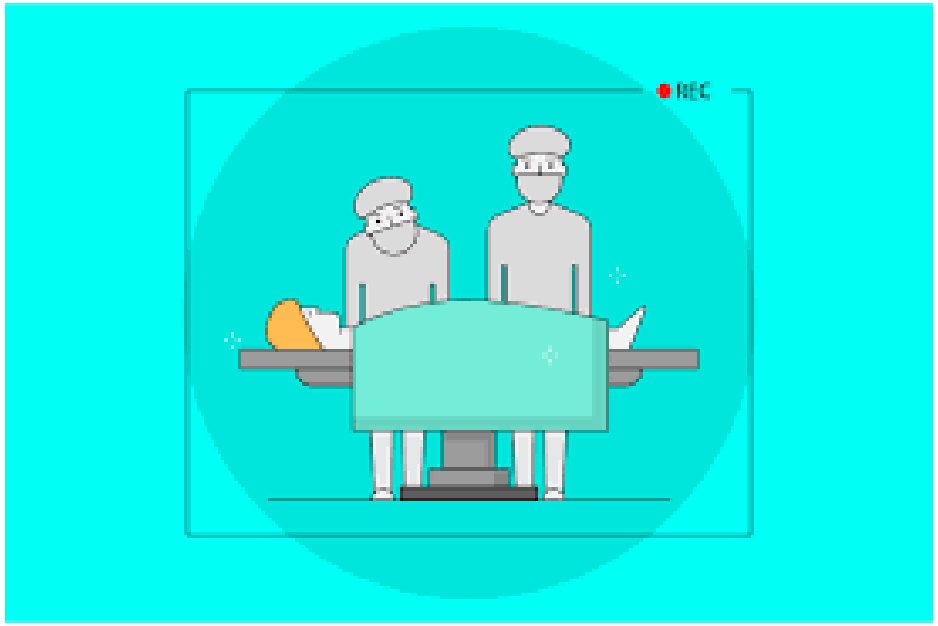


Image, video and sound capture - issues

Ethical issues in video recording – Medical ethics in neonatal units

Issues:

1. Informed Consent
2. Willingness
3. Privacy
4. Data Protection and Ownership



Src: https://www.jstor.org/stable/27720273?seq=1

Perpetuity Of Data Storage



BIG DATA
Mary Ann Liebert, Inc. publishers
Journals Search Alerts

Big Data. 2018 Sep 1; 6(3): 176–190.
Published online 2018 Sep 17. doi: 10.1089/big.2018.0083
PMCID: PMC6154451
PMID: 30283727

Aspects of Data Ethics in a Changing World: Where Are We Now?

David J. Hand*

[Aspects of Data Ethics in a Changing World: Where Are We Now? (nih.gov)]

A fundamental aspect of this is that one does not know, indeed *cannot* know, how data will be used in the future, or what other data they will be linked with. This means we cannot usefully characterize data sets as public (vs. not public) or by potential use (since these are unlimited and unforeseeable), and that the intrinsic nature of the data cannot be used as an argument that they are not risky. It is not the data *per se* that raise ethical issues, but the use to which they are put and the analysis to which they are subjected.

Perpetuity Of Data Storage

Consideration of 10 ethical issues:

1. The pervasiveness of modern data technology means that it can be legitimately regarded as part of **societal infrastructure**
2. The interconnectedness of data means that patterns can be detected beyond the (limited) data available
3. The dynamic nature of data means that that they may permit discoveries in the future that they do not permit today
4. Real-time and online analysis and decision-making, as data arrive. Is there a need for “circuit breakers”?
5. Synergistic analysis through merging and combination of data sets may change the nature of decisions and decision making

Perpetuity Of Data Storage

6. Lack of space, time, and social context limitation on scope of data as the gap between collection and analysis is extended.
7. Ability to use for unexpected purposes and to reveal unexpected information (role of data mining) may produce shocks
8. Risk of exceptional intrusiveness
9. Potential for misuse, privacy breach, blackmail, and other crimes.
10. Subtle ownership issues (whose data if I can sell it or share it?)

Perpetuity Of Data Storage/ Ethical Data Management Principles

High-level principles: integrity, honesty, objectivity, responsibility, trustworthiness, impartiality, nondiscrimination, transparency, accountability, fairness, robustness, resilience, usability, efficiency, and independence

Refined into lower principles for data science in government:

- start with clear user need and public benefit
- use data and tools that have the minimum intrusion necessary ("data minimization")
- create robust data science models (e.g., to avoid improper discrimination)
- be alert to public perceptions
- be as open and accountable as possible
- keep data secure



Questions?